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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

CS2104-1 – OVERVIEW OF OPERATING SYSTEMS

(For Minor Degree)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following is NOT a service provided by an operating system?
A) Program execution
B) Error detection
C) Data warehousing
D) I/O operations
2. Which type of system call is used to terminate a process?
A) fork ()
B) wait ()
C) exit ()
D) read ()
3. A linker is primarily used to:
A) Translate source code into machine code
B) Combine object code into a single executable
C) Execute programs on a CPU
D) Manage I/O operations
4. Which of the following is a type of user interface provided by an operating system?
A) Command Line Interface (CLI)
B) HTML Interface
C) Script Engine Interface
D) Media Player Interface
5. Which of the following defines a process?
A) A program in execution
B) A passive entity like a file
C) A memory management technique
D) A type of hardware
6. What is the primary function of a Process Control Block (PCB)?
A) Managing file I/O operations
B) Stores information about a process
C) Controls memory allocation
D) Controls CPU scheduling
7. In which state does a process use the CPU for execution?
A) Ready
B) Running
C) Waiting
D) New
8. What is the critical section in process synchronization?
A) A segment of code where shared resources are accessed
B) A method of process communication
C) A section of a program that never accesses shared data
D) A technique to prevent deadlock
9. Which of the following is true about Peterson's solution?
A) It uses semaphores for process synchronization
B) It allows multiple processes to enter the critical section simultaneously
C) It works for only two processes
D) It cannot guarantee mutual exclusion
10. Which of the following is a property of semaphores?
A) Semaphores can only be used for deadlock detection
B) Semaphores are a signaling mechanism
C) Semaphores do not ensure mutual exclusion
D) Semaphores are used to swap processes



11. What is a system model in deadlock characterization?
 - A) A method for deadlock prevention
 - B) A representation of processes and resources as a graph
 - C) A scheduling algorithm for processes
 - D) A technique for synchronization
12. Which of the following is a method for deadlock prevention?
 - A) Allow circular wait conditions
 - B) Ensure at least one process can acquire all resources
 - C) Allow processes to request resources only in a specific order
 - D) Make processes wait indefinitely
13. In memory management, what paging does?
 - A) Allocates entire memory to a single process
 - B) Divides the process memory into equal-sized pages
 - C) Preventing deadlocks from occurring
 - D) Ensures processes access memory in sequential order
14. What is a mutex lock?
 - A) A software-based synchronization technique
 - B) A hardware-based memory management technique
 - C) A type of semaphore
 - D) A scheduling algorithm
15. What is demand paging?
 - A) A paging system where pages are pre-loaded into memory.
 - B) A system where a page is loaded only when it is needed.
 - C) A method of dividing memory into fixed-size blocks.
 - D) A paging system that never requires swapping pages out of memory.
16. What is thrashing in the context of demand paging?
 - A) The process of repeatedly swapping pages in and out of memory, causing high overhead and slow performance.
 - B) A situation where all the pages are loaded in memory at the same time.
 - C) A memory allocation technique used to prevent page faults.
 - D) The process of dividing memory into multiple pages.
17. Which of the following page replacement algorithms is the most optimal, but not typically implementable in practice?
 - A) First-In, First-Out (FIFO)
 - B) Least Recently Used (LRU)
 - C) Optimal Page Replacement
 - D) Clock Page Replacement
18. Which of the following components of a file system structure contains information about the files and directories?
 - A) Superblock
 - B) Inode table
 - C) Data block
 - D) Boot block
19. Which of the following is a common file operation provided by most operating systems?
 - A) Process scheduling
 - B) File creation
 - C) Memory allocation
 - D) Network communication
20. In which directory implementation method are directory entries stored in a linked list, making it simple but slow for large directories?
 - A) Linear list
 - B) Tree-structured
 - C) Hash table
 - D) Indexed list

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the Operating system services with respect to programs and users.	8	L*2	1	2
	b) What are system calls? List and explain the different types of system call.	8	L2	1	2
2.	a) Define process. Explain different states of a process with state diagram.	8	L2	2	2
	b) Explain the various fields of PCB and explain the importance of each field.	8	L2	2	2

3. a) What is IPC? Explain direct and indirect communication with respect to message passing. 8 L2 2 2
- b) Analyze the performance of Operating system for the set of processes given in the table below using FCFS and SJF non-preemptive scheduling policy. Calculate average waiting time and average turnaround time.

Process	Arrival Time	Burst Time
P1	0	3
P2	2	3
P3	3	2
P4	5	5
P5	9	3

8 L3 2 2

Unit – II

4. a) Explaining the requirements must satisfy the critical section problem. 4 L2 3 2
- b) Illustrate with an example Petersons solution problem. 6 L3 3 2
- c) Explain paging with TLB with a neat diagram. 6 L2 3 2

5. a) Explain the different methods to recovery from the deadlock. 8 L2 4 2
- b) Consider the following snapshot of the system

Process	Allocation			Maximum			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

Find the need matrix and calculate the safe sequence using banker's algorithm-mention the above system is safe or not safe.

8 L3 4 2

6. a) Explain with diagram explain
- Paging
 - Swapping.
- b) Explain the structure of a page table with respect to hierarchical paging. 8 L2 4 2

Unit – III

7. a) With neat diagram, explain the concepts of demand paging. 8 L2 5 2
- b) Find the number of page faults for the following page reference string using FIFO and LRU policies with four free frames. Which algorithm gives the minimum number of page faults?

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

8 L3 5 2

8. a) What is a file? What are its attributes? Explain file operations. 8 L2 5 2
- b) Discuss the various methods to implement the free space list. 8 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
